

Instructions:

- This is an optional extra credit assignment. It will be weighted as an additional problem set in the final grade if completed.
- Submissions are due by 9:00 p.m. Thursday 7/23/2026. Solutions will be released at this time.
- We strongly recommend attempting these problems alone first to help gauge your level of understanding for review.

Problem 1. 10 pts: In class we discussed four categories of focusing: quadrupole (strong) focusing, solenoid focusing, bending dipole focusing, and dipole edge focusing.

- a) Discuss the differences between solenoid and quadrupole focusing. What is achieved in each case? When might each be used?
- b) Assume you have a circular accelerator in the horizontal plane with a mixture of rectangular and sector bend dipoles. In which magnets do the two types of dipole focusing occur, and which transverse dimension do they affect?

Problem 2. 10 pts: Recall Problem 3 from the midterm. Assume the bunch that is injected has an elliptical phase space with no average correlation in time and energy. Sketch the phase at injection, $\tau/2$, and τ for two adjacent bunches where τ is the debunching time.

Problem 3. 10 pts: In class, we have discussed the Courant-Snyder Invariant

$$C(u, u') = \frac{1}{\beta} \left[u^2 + (\alpha u + \beta u')^2 \right] = \gamma u^2 + 2\alpha u u' + \beta u'^2$$

where u is an arbitrary transverse coordinate.

- a) Explain in words the meaning of the Courant-Snyder Invariant. When is the Courant-Snyder Invariant actually invariant?
- b) Briefly explain the difference between emittance, RMS emittance, and normalized emittance.

Problem 4. 10 pts: Use three wires to create a field at $x=0$ and $y=0$ that only produces magnetic field in the $\hat{y}$ direction, and the magnitude of B_y is $3\mu_0 I_0 / (\pi a)$. The wires are carrying a current I_0 each. Keep the wires far apart and the wires cannot overlap.

Problem 5. 10 pts: Circular Colliders

- a) If you want to reduce the radiation losses (power) at 100 GeV for LEP by a factor of 10 how big should the e^+e^- bending radius be?
- b) What would happen if you went from electrons to muons and kept the same kinetic energy?